

SOME THEOREMS ON QUADRATIC FORMS AND NORMAL VARIABLES

1. THE MULTIVARIATE NORMAL DISTRIBUTION

The $n \times 1$ vector of random variables, y , is said to be distributed as a multivariate normal with mean vector μ and variance covariance matrix Σ (denoted $y \sim N(\mu, \Sigma)$) if the density of y is given by

$$f(y; \mu, \Sigma) = \frac{e^{-\frac{1}{2}(y-\mu)'\Sigma^{-1}(y-\mu)}}{(2\pi)^{\frac{n}{2}}|\Sigma|^{\frac{1}{2}}} \quad (1)$$

Consider the special case where $n = 1$: $y = y_1, \mu = \mu_1, \Sigma = \sigma^2$.

$$\begin{aligned} f(y_1; \mu_1, \sigma) &= \frac{e^{-\frac{1}{2}(y_1-\mu_1)\left(\frac{1}{\sigma^2}\right)(y_1-\mu_1)}}{(2\pi)^{\frac{1}{2}}(\sigma^2)^{\frac{1}{2}}} \\ &= \frac{e^{-\frac{(y_1-\mu_1)^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \end{aligned} \quad (2)$$

is just the normal density for a single random variable.

2. THEOREMS ON QUADRATIC FORMS IN NORMAL VARIABLES

2.1. Quadratic Form Theorem 1.

Theorem 1. *If $y \sim N(\mu_y, \Sigma_y)$, then*

$$z = Ay \sim N(\mu_z = A\mu_y; \Sigma_z = A\Sigma_y A')$$

where A is a matrix of constants.

2.1.1. *Proof.*

$$\begin{aligned} E(z) &= E(Ay) = AE(y) = A\mu_y \\ \text{var}(z) &= E[(z - E(z))(z - E(z))'] \\ &= E[(Ay - A\mu_y)(Ay - A\mu_y)'] \\ &= E[A(y - \mu_y)(y - \mu_y)'A'] \\ &= AE(y - \mu_y)(y - \mu_y)'A' \\ &= A\Sigma_y A' \end{aligned} \quad (3)$$

2.1.2. *Example.* Let $Y_1, \dots, Y_n$ denote a random sample drawn from $N(\mu, \sigma^2)$. Then

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} \sim N \left[\begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix}, \begin{pmatrix} \sigma^2 & \cdots & 0 \\ \vdots & \sigma^2 & \vdots \\ 0 & & \sigma^2 \end{pmatrix} \right] \quad (4)$$

Now Theorem 1 implies that:

$$\begin{aligned} \bar{Y} &= \frac{1}{n}Y_1 + \cdots + \frac{1}{n}Y_n \\ &= \left(\frac{1}{n}, \dots, \frac{1}{n} \right) Y = AY \\ &\sim N(\mu, \sigma^2/n) \quad \text{since} \\ &\left(\frac{1}{n}, \dots, \frac{1}{n} \right) \begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix} = \mu \quad \text{and} \end{aligned} \quad (5)$$

$$\left(\frac{1}{n}, \dots, \frac{1}{n} \right) \sigma^2 I \begin{pmatrix} \frac{1}{n} \\ \vdots \\ \frac{1}{n} \end{pmatrix} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

2.2. Quadratic Form Theorem 2.

Theorem 2. Let the $n \times 1$ vector $y \sim N(0, I)$. Then $y'y \sim \chi^2(n)$.

Proof: Consider that each y_i is an independent standard normal variable. Write out $y'y$ in summation notation as

$$y'y = \sum_{i=1}^n y_i^2 \quad (6)$$

which is the sum of squares of n standard normal variables.

2.3. Quadratic Form Theorem 3.

Theorem 3. If $y \sim N(0, \sigma^2 I)$ and M is a symmetric idempotent matrix of rank m then

$$\frac{y'My}{\sigma^2} \sim \chi^2(\text{tr } M) \quad (7)$$

Proof: Since M is symmetric it can be diagonalized with an orthogonal matrix Q . This means that

$$Q'MQ = \Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix} \quad (8)$$

Furthermore, since M is idempotent all these roots are either zero or one. Thus we can choose Q so that Λ will look like

$$Q'MQ = \Lambda = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} \quad (9)$$

The dimension of the identity matrix will be equal to the rank of M , since the number of non-zero roots is the rank of the matrix. Since the sum of the roots is equal to the trace, the dimension is also equal to the trace of M . Now let $v = Q'y$. Compute the moments of

$$\begin{aligned} v &= Q'y \\ E(v) &= Q'E(y) = 0 \\ \text{var}(v) &= Q'\sigma^2 IQ \\ &= \sigma^2 Q'Q = \sigma^2 I \quad \text{since } Q \text{ is orthogonal} \\ \Rightarrow v &\sim N(0, \sigma^2 I) \end{aligned} \quad (10)$$

Now consider the distribution of $y'My$ using the transformation v . Since Q is orthogonal, its inverse is equal to its transpose. This means that $y = (Q')^{-1}v = Qv$. Now write the quadratic form as follows

$$\begin{aligned} \frac{y'My}{\sigma^2} &= \frac{v'Q'MQv}{\sigma^2} \\ &= \frac{1}{\sigma^2} v' \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} v \\ &= \frac{1}{\sigma^2} \sum_{i=1}^{\text{tr } M} v_i^2 \\ &= \sum_{i=1}^{\text{tr } M} \left(\frac{v_i}{\sigma} \right)^2 \end{aligned} \quad (11)$$

This is the sum of squares of $(\text{tr } M)$ standard normal variables and so is a χ^2 variable with $\text{tr } M$ degrees of freedom.

Corollary: If the $n \times 1$ vector $y \sim N(0, I)$ and the $n \times n$ matrix A is idempotent and of rank m . Then

$$y' Ay \sim \chi^2(m)$$

2.4. Quadratic Form Theorem 4.

Theorem 4. If $y \sim N(0, \sigma^2 I)$, M is a symmetric idempotent matrix of order n , and L is a $k \times n$ matrix, then Ly and $y' My$ are independently distributed if $LM = 0$.

Proof: Define the matrix Q as before so that

$$Q' MQ = \Lambda = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} \quad (12)$$

Let r denote the dimension of the identity matrix which is equal to the rank of M . Thus $r = \text{tr } M$.

Let $v = Q'y$ and partition v as follows

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_r \\ v_{r+1} \\ \vdots \\ v_n \end{bmatrix} \quad (13)$$

The number of elements of v_1 is r , while v_2 contains $n - r$ elements. Clearly v_1 and v_2 are independent of each other since they are independent standard normals. What we will show now is that $y' My$ depends only on v_1 and Ly depends only on v_2 . Given that the v_i are independent, $y' My$ and Ly will be independent. First use Theorem 3 to note that

$$\begin{aligned} y' My &= v' Q' MQ v \\ &= v' \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} v \\ &= v_1' v_1 \end{aligned} \quad (14)$$

Now consider the product of L and Q which we denote C . Partition C as (C_1, C_2) . C_1 has k rows and r columns. C_2 has k rows and $n - r$ columns. Now consider the following product

$$\begin{aligned} C(Q' MQ) &= LQQ' MQ, \text{ since } C = LQ \\ &= LMQ = 0, \text{ since } LM = 0 \text{ by assumption} \end{aligned} \quad (15)$$

Now consider the product of C and the matrix $Q'MQ$

$$\begin{aligned} C(Q'MQ) &= (C_1, C_2) \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} \\ &= 0 \end{aligned} \tag{16}$$

This of course implies that $C_1 = 0$. This then implies that

$$LQ = C = (0, C_2) \tag{17}$$

Now consider Ly . It can be written as

$$\begin{aligned} Ly &= LQQ'y, \text{ since } Q \text{ is orthogonal} \\ &= Cv, \text{ by definition of } C \text{ and } v \\ &= C_2v_2, \text{ since } C_1 = 0 \end{aligned} \tag{18}$$

Now note that Ly depends only on v_2 , and $y'My$ depends only on v_1 . But since v_1 and v_2 are independent, so are Ly and $y'My$.

2.5. Quadratic Form Theorem 5.

Theorem 5. *Let the $n \times 1$ vector $y \sim N(0, I)$, let A be an $n \times n$ idempotent matrix of rank m , let B be an $n \times n$ idempotent matrix of rank s , and suppose $BA = 0$. Then $y'Ay$ and $y'By$ are independently distributed χ^2 variables.*

Proof: By Theorem 3 both quadratic forms are distributed as chi-square variables. We need only to demonstrate their independence. Define the matrix Q as before so that

$$Q'AQ = \Lambda = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \tag{19}$$

Let $v = Q'y$ and partition v as

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_r \\ v_{r+1} \\ \vdots \\ v_n \end{bmatrix} \tag{20}$$

Now form the quadratic form $y'Ay$ and note that

$$\begin{aligned} y'Ay &= v'Q'AQv \\ &= v' \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} v \\ &= v_1'v_1 \end{aligned} \tag{21}$$

Now define $G = Q'BQ$. Since B is only considered as part of a quadratic form we may consider that it is symmetric, and thus note that G is also symmetric. Now form the product $G\Lambda = Q'BQQ'AQ$. Since Q is orthogonal its transpose is equal to its inverse and we can write $G\Lambda = Q'BAQ = 0$, since $BA = 0$ by assumption. Now write out this identity in partitioned form as

$$\begin{aligned} G(Q'AQ) &= \begin{pmatrix} G_1 & G_2 \\ G_2' & G_3 \end{pmatrix} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \\ &= \begin{pmatrix} G_1 & 0 \\ G_2' & 0 \end{pmatrix} = \begin{bmatrix} 0_r & 0 \\ 0 & 0 \end{bmatrix} \end{aligned} \quad (22)$$

where G_1 is $r \times r$, G_2 is $r \times (n-r)$ and G_3 is $(n-r) \times (n-r)$.

This means then that $G_1 = 0_r$ and $G_2 = G_2' = 0$.

This means that G is given by

$$G = \begin{pmatrix} 0 & 0 \\ 0 & G_3 \end{pmatrix} \quad (23)$$

Given this information write the quadratic form in B as

$$\begin{aligned} y'By &= y'Q'QBQQ'y \\ &= v'Gv \\ &= (v_1', v_2') \begin{bmatrix} 0 & 0 \\ 0 & G_3 \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\ &= v_2'G_3v_2 \end{aligned} \quad (24)$$

It is now obvious that $y'Ay$ can be written in terms of the first r terms of v , while $y'By$ can be written in terms of the last $n-r$ terms of v . Since the v 's are independent the result follows.

2.6. Quadratic Form Theorem 6 (Craig's Theorem).

Theorem 6. *If $y \sim N(\mu, \Omega)$ where Ω is positive definite, then $q_1 = y'Ay$ and $q_2 = y'By$ are independently distributed if $A\Omega B = 0$.*

Proof of sufficiency:

This is just a generalization of Theorem 5. Since Ω is a covariance matrix of full rank it is positive definite and can be factored as $\Omega = TT'$. Therefore the condition $A\Omega B = 0$ can be written $ATT'B = 0$. Now pre-multiply this expression by T' and post-multiply by T to obtain that $T'ATT'BT = 0$. Now define $C = T'AT$ and $K = T'BT$ and note that if $A\Omega B = 0$, then

$$CK = (T'AT)(T'BT) = T'\Omega BT = T'0T = 0 \quad (25)$$

Consequently, due to the symmetry of C and K , we also have

$$0 = 0' = (CK)' = K'C' = KC \quad (26)$$

Thus $CK = 0$ and $KC = 0$ and $KC = CK$. A simultaneous diagonalization theorem in matrix algebra [9, Theorem 4.15, p. 155] says that if $CK = KC$ then there exists an orthogonal matrix Q such that

$$Q'CQ = \begin{bmatrix} D_1 & 0 \\ 0 & 0 \end{bmatrix} \quad (27)$$

$$Q'KQ = \begin{bmatrix} 0 & 0 \\ 0 & D_2 \end{bmatrix}$$

where D_1 is an $n_1 \times n_1$ diagonal matrix and D_2 is an $(n - n_1) \times (n - n_1)$ diagonal matrix. Now define $v = Q'T^{-1}y$. It is then distributed as a normal variable with expected value and variance given by

$$\begin{aligned} E(v) &= Q'T^{-1}\mu \\ \text{var}(v) &= Q'T^{-1}\Omega T^{-1'}Q \\ &= Q'T^{-1}TT'T^{-1'}Q \\ &= I \end{aligned} \quad (28)$$

Thus the vector v is a vector of independent standard normal variables.

Now consider $q_1 = y'Ay$ in terms of v . First note that $y = TQv$ and that $y' = v'Q'T'$. Now write out $y'Ay$ as follows

$$\begin{aligned} q_1 &= y'Ay = v'Q'T'ATQv \\ &= v'Q'T'(T'^{-1}CT^{-1})TQv \\ &= v'Q'CQv \\ &= v_1'D_1v_1 \end{aligned} \quad (29)$$

Similarly we can define $y'By$ in terms of v as

$$\begin{aligned} q_2 &= y'By = v'Q'T'BTQv \\ &= v'Q'T'(T'^{-1}KT^{-1})TQv \\ &= v'Q'KQv \\ &= v_2'D_2v_2 \end{aligned} \quad (30)$$

Thus $q_1 = y'Ay$ is defined in terms of the first n_1 elements of v , and $q_2 = y'By$ is defined in terms of the last $n - n_1$ elements of v and so they are independent.

The proof of necessity is difficult and has a long history [2], [3].

2.7. Quadratic Form Theorem 7.

Theorem 7. *If y is a $n \times 1$ random variable and $y \sim N(\mu, \Sigma)$ then*

$$(y - \mu)' \Sigma^{-1} (y - \mu) \sim \chi^2(n)$$

Proof: Let $w = (y - \mu)' \Sigma^{-1} (y - \mu)$. If we can show that $w = z'z$ where z is distributed as $N(0, I)$ then the proof is complete. Start by diagonalizing Σ with an orthogonal matrix Q . Since Σ is positive definite all the elements of the diagonal matrix Λ will be positive.

$$Q' \Sigma Q = \Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix} \quad (31)$$

Now let Λ^* be the following matrix defined based on Λ .

$$\Lambda^* = \begin{bmatrix} \frac{1}{\sqrt{\lambda_1}} & 0 & 0 & \dots & 0 \\ 0 & \frac{1}{\sqrt{\lambda_2}} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & \frac{1}{\sqrt{\lambda_n}} \end{bmatrix} \quad (32)$$

Now let the matrix $H = Q' \Lambda^* Q$. Obviously H is symmetric. Furthermore

$$\begin{aligned} H' H &= Q' \Lambda^* Q Q' \Lambda^* Q \\ &= Q' \Lambda^{-1} Q \\ &= \Sigma^{-1} \end{aligned} \quad (33)$$

The last equality follows from the definition of $\Sigma = Q \Lambda Q'$ after taking the inverse of both sides remembering that the inverse of an orthogonal matrix is equal to its transpose. Furthermore it is obvious that

$$\begin{aligned} H \Sigma H' &= Q \Lambda^* Q' \Sigma Q \Lambda^* Q' \\ &= Q \Lambda^* Q' Q \Lambda Q' Q \Lambda^* Q' \\ &= I \end{aligned} \quad (34)$$

Now let $\varepsilon = y - \mu$ so that $\varepsilon \sim N(0, \Sigma)$. Now consider the distribution of $z = H\varepsilon$. It is a standard normal since

$$\begin{aligned}
E(z) &= HE(\varepsilon) = 0 \\
\text{var}(z) &= H \text{var}(\varepsilon) H' \\
&= H \Sigma H' \\
&= I
\end{aligned} \tag{35}$$

Now write w as $w = \varepsilon \Sigma^{-1} \varepsilon$ and see that it is equal to $z'z$ as follows

$$\begin{aligned}
w &= \varepsilon' \Sigma^{-1} \varepsilon \\
&= \varepsilon' H' H \varepsilon \\
&= (H \varepsilon)' (H \varepsilon) \\
&= z' z
\end{aligned} \tag{36}$$

2.8. Quadratic Form Theorem 8. Let $y \sim N(0, I)$. Let M be a non-random idempotent matrix of dimension $n \times n$ ($\text{rank}(M) = r \leq n$). Let A be a non-random matrix such that $AM = 0$. Let $t_1 = My$ and let $t_2 = Ay$. Then t_1 and t_2 are independent random vectors.

Proof: Since M is symmetric and idempotent it can be diagonalized using an orthonormal matrix Q as before.

$$Q' M Q = \Lambda = \begin{bmatrix} I_{r \times r} & 0_{r \times (n-r)} \\ 0_{(n-r) \times r} & 0_{(n-r) \times (n-r)} \end{bmatrix} \tag{37}$$

Further note that since Q is orthogonal that $M = Q \Lambda Q'$. Now partition Q as $Q = (Q_1, Q_2)$ where Q_1 is $n \times r$. Now use the fact that Q is orthonormal to obtain the following identities

$$\begin{aligned}
Q Q' &= (Q_1 Q_2) \begin{pmatrix} Q_1' \\ Q_2' \end{pmatrix} \\
&= Q_1 Q_1' + Q_2 Q_2' = I_n
\end{aligned} \tag{38}$$

$$\begin{aligned}
Q' Q &= \begin{pmatrix} Q_1' \\ Q_2' \end{pmatrix} (Q_1 Q_2) = \begin{bmatrix} Q_1' Q_1 & Q_1' Q_2 \\ Q_2' Q_1 & Q_2' Q_2 \end{bmatrix} \\
&= \begin{pmatrix} I_r & 0 \\ 0 & I_{n-r} \end{pmatrix}
\end{aligned}$$

Now multiply Λ by Q to obtain

$$\begin{aligned}
Q \Lambda &= (Q_1 Q_2) \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix} \\
&= (Q_1 \ 0)
\end{aligned} \tag{39}$$

Now compute M as

$$\begin{aligned} M &= Q\Lambda Q' = (Q_1 Q_2) \begin{pmatrix} Q'_1 \\ Q'_2 \end{pmatrix} \\ &= Q_1 Q'_1 \end{aligned} \tag{40}$$

Now let $z_1 = Q'_1 y$ and let $z_2 = Q'_2 y$. Note that

$$z = (z'_1, z'_2) = C' y$$

is a standard normal since $E(x) = 0$ and $\text{var}(z) = CC' = I$. Furthermore z_1 and z_2 are independent. Now consider $t_1 = My$. Rewrite this using (40) as

$$Q_1 Q'_1 y = Q_1 z_1$$

Thus t_1 depends only on z_1 . Now let the matrix

$$N = I - M = Q_2 Q'_2$$

from (38) and (40). Now notice that

$$AN = A(I - M) = A - AM = A$$

since $AM = 0$. Now consider $t_2 = Ay$. Replace A with AN to obtain

$$\begin{aligned} t_2 &= Ay = ANy \\ &= A(Q_2 Q'_2)y \\ &= AQ_2(Q'_2 y) \\ &= AQ_2 z_2 \end{aligned} \tag{41}$$

Now t_1 depends only on z_1 and t_2 depends only on z_2 and since the z s are independent the t s are also independent.

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